Lorentzian Einstein–Cartan Reduction on Four Homogeneous Geometries: Curvature, Boundary Terms, and Reduced Orbits

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Abstract

We compare Lorentzian Einstein–Cartan gravity with constant complete Nieh–Yan coupling on fixed compact representatives of T^3, Nil, Sol and S^3. In the common-scale, two-amplitude family, torsion is auxiliary and χ organizes reduced trajectories and specified curvature scalars. The full connection equation and five-direction shape variation restrict real common-scale vacuum families to locally flat static ones. On the flat torus, the second-order obstruction to pure scale extends to a mixed scale–shear constraint cone; each direction is realized by an exact local vacuum family. For Nil and Sol, initial tracefree Ricci determines quadratic shape evolution and the quartic scale departure. Selected quadratic and cubic curvature contractions require J_2, J_3 in addition to χ, recovering the spatial Ricci spectrum at a point. We derive degenerate branches of the inverse structure-matrix problem, global frames of the same flat torus and canonical endpoint correspondence, distinguishing reduced trajectories, full vacuum solutions, structure representations and boundary data.

1 Introduction

Homogeneous reduction of Lorentzian Einstein–Cartan gravity involves several related objects: local curvature contractions, spatial integrals, trajectories of the reduced action, and solutions of the full field equations. We compare the fixed reference geometries T^3, Nil, Sol and S^3, asking which results are controlled by their scalar-curvature parameter χ. The model has no curvature-square term, matter source or cosmological constant, and its complete Nieh–Yan (NY) coupling is constant [7, 8].

The common-scale family consists of a fixed spatial metric with a constant class-A structure matrix, one time-dependent scale and two geometric torsion amplitudes. After eliminating torsion, we also vary all five homogeneous tracefree shape components. Nil and Sol have anisotropic spatial curvature. A common scale multiplies all spatial directions by the same factor; it does not make their metrics isotropic. The quantity χ describes a normalized reference metric, not a general topological invariant.

Earlier papers study reduced potentials, curvature identities, homogeneous deformations and torsion variations [1, 2, 3, 4]. Paper04 treats structure matrices, quadratic forms and boundary data on four fixed compact quotients [5]; Paper05 develops the Lorentzian action, endpoint terms and homogeneous variations on the sphere normalized by its physical radius [6]. Here we derive curvature formulas for a constant class-A structure matrix and compare the four reduced orbit problems with the full field equations.

1.1 Main results

After separating complete NY into an endpoint term, both torsion amplitudes are auxiliary and vanish uniquely. The remaining lapse constraint classifies real reduced trajectories by the sign of χ: the spherical constraint surface is empty, the torus has a positive constant scale, and Nil and Sol have two linear proper-time trajectories. The latter retain a nonzero spatial tracefree Einstein residual and therefore are not full source-free vacuum solutions.

We derive NY, Pontryagin, a covariant internal-dual contraction and a relative Pontryagin primitive for arbitrary time profiles. Within this family their local scalar expressions depend on the structure matrix through χ, and the static mixed-torsion coefficient is $C = -9\chi$. Absolute Chern–Simons (CS) primitives, spatial tracefree Ricci curvature and the volume normalization of integrals describe the geometric information needed in addition to scalar curvature.

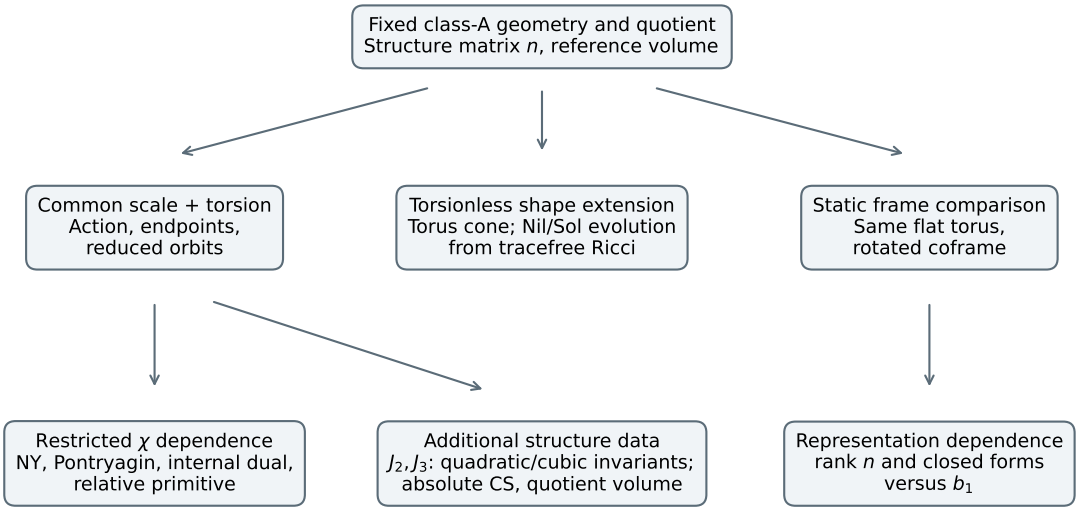


Figure 1: Organization of the comparison. The common-scale family with torsion, the torsionless shape extension and the static frame comparison have distinct domains. Within the common-scale family, quantities controlled by χ are separated from contractions that need J_2, J_3 or quotient data.

Section 2 fixes the geometry. Sections 3–4 derive the action, endpoints and reduced trajectories. Section 5 derives common-scale conditions and shape evolution; Section 6 gives time-dependent curvature, selected invariants and relative endpoints; Section 7 analyzes curvature information and its limits. The appendices supply convention maps, curvature identities, variational details and reproduction instructions.

1.2 Scope of the comparison

Two questions organize the comparison: the validity and shape extension of the common-scale reduction, and the curvature information retained by χ, J_2, J_3. On the torus, pure scale has a second-order obstruction, whereas each direction of a mixed scale–shear cone is realized by an exact vacuum family. For Nil and Sol, the residual determines the initial shape acceleration and the quartic scale coefficient.

The curvature calculation connects to left-invariant metric geometry [11]; the exact family uses the Kasner relations [12]. Integrability of linearized Einstein solutions has a broader history: the closed flat torus is the standard example of linearization instability [14], and a vacuum

spacetime with a compact constant-mean-curvature Cauchy surface and Killing fields is a conical point of the solution space [13, 15]. Here necessity and sufficiency are proved directly in a finite homogeneous sector on a fixed lattice, without invoking a general linearization-stability theorem. Section 5 and Appendix D treat shape; Sections 6–7 and Appendix B treat invariants and the inverse problem.

2 Geometry and variation domain

Use an oriented orthonormal coframe. The real symmetric spatial structure matrix n is time independent and can be diagonalized by a constant $SO(3)$ frame rotation. Spatial indices run over $1, 2, 3$ and spacetime indices over $0, 1, 2, 3$. Repeated spatial indices in the structure equation are summed.

$$\begin{aligned} &g_{ab} = \mathrm{diag}(-1, 1, 1, 1), \qquad \epsilon_{0123} = +1, \quad \epsilon^{0123} = -1, \\ &e^0 = N dt, \qquad e^i = q\sigma^i, \qquad d\sigma^i = \tfrac{1}{2} n_{ij}\epsilon_{jkl}\sigma^k \wedge \sigma^l, \\ &D = N^{-1} d/dt, \qquad H = Dq/q, \qquad \mathcal{V}_\Sigma = \int_\Sigma \sigma^{123}, \quad q, N > 0. \end{aligned} \tag{G-01}$$

This class-A representation satisfies the Jacobi identity. In a diagonal frame let n_i be the eigenvalues and k_i the sectional-curvature coefficient in the complementary spatial plane. The ith spatial Ricci eigenvalue is the sum of the other two k_i. The scalar-curvature parameter is defined by the resulting curvature.

$$\begin{aligned} \ell_i &= \tfrac{1}{2}(n_1 + n_2 + n_3) - n_i, \qquad k_i = n_i\ell_i - \ell_j\ell_k, \\ {}^{(3)}R &= \frac{2(k_1 + k_2 + k_3)}{q^2} = \frac{6\chi}{q^2}, \\ 6\chi &= \tfrac{1}{2}(\mathrm{tr}\, n)^2 - \mathrm{tr}(n^2), \qquad {}^{(3)}\mathrm{Ric}_{ii} = \frac{k_j + k_k}{q^2}. \end{aligned} \tag{G-02}$$

In (G-02), (i, j, k) denotes each cyclic triple, with no further summation. The four reference geometries are fixed below. The spherical q is the physical round radius, as in Paper05.

Geometry	Eigenvalues of n	χ	Reference volume $\mathcal{V}_\Sigma$
T3	$(0, 0, 0)$	0	$(2\pi)^3$
Nil	$(0, 0, -1)$	$-1/12$	$(2\pi)^4$
Sol	$(0, 1, -1)$	$-1/3$	$\log((3 + \sqrt{5})/2)$
S3	$(2, 2, 2)$	1	$2\pi^2$

On the torus all three angular coordinates have period 2π. Spatial coordinates are written x^1, x^2, x^3. For Nil take the coframe $dx^1, dx^2, dx^3 - x^1 dx^2$ and the lattice generators $(x^1, x^2, x^3) \mapsto (x^1 + 2\pi, x^2, x^3 + 2\pi x^2)$, $(x^1, x^2, x^3) \mapsto (x^1, x^2 + 2\pi, x^3)$ and $(x^1, x^2, x^3) \mapsto (x^1, x^2, x^3 + 4\pi^2)$. Each identification preserves the coframe.

For Sol use $dx^1, e^{x^1} dx^2, e^{-x^1} dx^3$. Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, with eigenvalues $\lambda = (3 + \sqrt{5})/2$ and λ^{-1}. Identify a shift of x^1 by $\log \lambda$ with the fiber map $\mathrm{diag}(\lambda^{-1}, \lambda)$, using the area-one integer lattice transported to eigen-coordinates. The coframe descends to this mapping torus. Before diagonalization, $n_{23} = n_{32} = 1$ are its only nonzero entries. This is class A, type VI_0 [5].

Define geometric torsion by the first Cartan equation and fix its two real amplitudes by the following ansatz. Again, the spatial wedge is displayed once for each cyclic triple.

$$T^a = de^a + \omega^a{}_b \wedge e^b, \qquad T^0 = 0, \qquad T^i = \frac{V}{3} e^{0i} + \frac{2\eta}{q} e^{jk}. \tag{G-03}$$

$$
\begin{aligned}
a_i &= (\ell_i - \eta)/q, \qquad b = H - V/3, \\
\omega^0{}_i &= \omega^i{}_0 = be^i, \qquad \omega^j{}_k = a_i e^i, \qquad \omega^k{}_j = -a_i e^i.
\end{aligned}
\tag{G-04}
$$

Substitution of (G-04) into (G-03) reproduces the specified torsion. Profiles are sufficiently smooth, while the spatial shape and quotient lattice are fixed. Formulas for a general structure matrix are local identities; the explicit spatial integrals use the fixed compact quotients above. Section 5 tests how equations obtained after this restriction relate to the full independent connection and coframe equations.

2.1 Distinct variation domains

The common-scale, two-amplitude family uses fixed real symmetric n and q, N, η, V; this is the domain of the time-dependent curvature formulas in Section 6. For shape, we first solve the independent connection equation and use the LC action with $e^i = q(e^{\mathsf{B}})^i{}_j \sigma^j$, $\mathsf{B}^T = \mathsf{B}$, $\mathrm{tr}\mathsf{B} = 0$. The five-component matrix B differs from the endpoint generator B_{end}. The physical structure matrix is $m = q^{-1} e^{\mathsf{B}} n e^{\mathsf{B}}$.

A positive definite constant coefficient metric in the fixed global coframe descends to the same quotient. A constant congruence need not preserve the lattice and is not identified with a coordinate gauge transformation. The rotated coframe in Section 7 and Appendix A compares representations of the same static flat torus metric; it does not identify the entire dynamical constant-shape-coefficient families of the two frames.

3 Bulk action and signed boundary completion

Use the curvature of the same connection and include complete NY as an exterior derivative. The coupling θ_{NY} is constant and $\kappa^2 > 0$. The boundaries are spacelike endpoint slices; the spatial quotient is closed.

$$
\begin{aligned}
S &= \int_M e^{0123} \frac{R_{\mathrm{EC}}}{2\kappa^2} + \theta_{\mathrm{NY}} \int_M dQ + S_{\mathrm{GHY}}, \\
R^a{}_b &= d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b, \qquad Q = e^a \wedge T_a, \\
dQ &= T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}.
\end{aligned}
\tag{B-01}
$$

Differentiating the time-dependent connection gives the following scalar-curvature decomposition. The derivative of the vector amplitude and its product with H form one divergence after multiplication by the volume density.

$$
\begin{aligned}
R_{\mathrm{LC}} &= 6(DH + 2H^2 + \chi/q^2), \\
R_{\mathrm{EC}} - R_{\mathrm{LC}} &= -6\eta^2/q^2 + 2V^2/3 - 2DV - 6HV.
\end{aligned}
\tag{B-02}
$$

$$
\begin{aligned}
L_{\mathrm{raw}} &= L_0 + \frac{d}{dt}(B_{\mathrm{EH}} + B_{\mathrm{EC}} + B_{\mathrm{NY}}), \\
L_0 &= A_\Sigma q \left(NF - \frac{\dot{q}^2}{N} \right), \qquad A_\Sigma = \frac{3\mathcal{V}_\Sigma}{\kappa^2}, \\
F &= \chi - \eta^2 + \frac{q^2 V^2}{9}.
\end{aligned}
\tag{B-03}
$$

$$
B_{\mathrm{EH}} = \frac{3\mathcal{V}_\Sigma q^2 \dot{q}}{\kappa^2 N}, \qquad B_{\mathrm{EC}} = -\frac{\mathcal{V}_\Sigma q^3 V}{\kappa^2}, \qquad B_{\mathrm{NY}} = 6\theta_{\mathrm{NY}} \mathcal{V}_\Sigma q^2 \eta. \tag{B-04}
$$

Equation (B-03) is common to the four geometries. The structure matrix enters F through χ, while the quotient volume enters A_Σ. Complete NY is contained in $B_{\rm NY}$, so the bulk auxiliary equations and reduced trajectories are independent of its constant coupling.

The outward unit normal points to the future at the final slice and to the past at the initial slice. Use the LC extrinsic curvature and the signature factor in the GHY term [9, 10].

$$\begin{aligned} &\hat{n}_f = +N^{-1}\partial_t, \quad \hat{n}_i = -N^{-1}\partial_t, \quad \epsilon_n = \hat{n}^a\hat{n}_a = -1, \quad K_f = 3H_f, \quad K_i = -3H_i, \\ &S_{\rm GHY} = \frac{1}{\kappa^2}\sum_{s=i,f}\epsilon_n\int_{\Sigma_s}\sqrt{h}K_s = -[B_{\rm EH}]_i^f. \end{aligned} \tag{B-05}$$

$$S = \int_{t_i}^{t_f} L_0 dt + [B_{\rm EC} + B_{\rm NY}]_i^f, \qquad \delta q = \delta\eta = \delta V = 0 \quad \text{at both endpoints.} \tag{B-06}$$

These fixed endpoint data suffice for the completed variational problem; neither the endpoint lapse nor the scale velocity needs to be fixed. If L_0 is chosen as the canonical representative, restoring the derivative of $B = B_{\rm EC} + B_{\rm NY}$ shifts the momenta as follows, without changing the bulk equations.

$$\begin{aligned} &\Delta p_q = -\frac{3\mathcal{V}_\Sigma q^2 V}{\kappa^2} + 12\theta_{\rm NY}\mathcal{V}_\Sigma q\eta, \\ &\Delta p_\eta = 6\theta_{\rm NY}\mathcal{V}_\Sigma q^2, \qquad \Delta p_V = -\frac{\mathcal{V}_\Sigma q^3}{\kappa^2}, \qquad \Delta p_N = 0. \end{aligned} \tag{B-07}$$

Prescribing nonzero endpoint torsion is distinct from the existence of a smooth stationary curve with those endpoint values. The auxiliary equations in Section 4 impose $\eta = V = 0$ throughout the interval and, by continuity, at its endpoints. Boundary sources or discontinuous layers define different problems and are not included here.

3.1 Canonical correspondence of endpoint representatives

After GHY completion, write $B_{\rm end} = B_{\rm EC} + B_{\rm NY}$. Adding $dB_{\rm end}/dt$ gives $P_i = p_i + \partial_i B_{\rm end}$, including the auxiliary momenta. In $\widetilde{p}_i = P_i - \partial_i B_{\rm end}$, the constraints retain their original form. Appendix C gives the two-form and second-class constraints. Canonical equivalence refers to corresponding endpoint data: fixing the old and new momenta independently would generally define different boundary conditions.

4 Auxiliary constraints and reduced trajectories

Vary L_0 before selecting a time gauge, with Poisson convention $\{q, p_q\} = 1$. The canonical variables and Hamiltonian constraint are

$$\begin{aligned} &p_q = -2A_\Sigma q\dot{q}/N, \qquad p_N = p_\eta = p_V = 0, \\ &H_0 = N\mathcal{C}, \qquad \mathcal{C} = -\frac{p_q^2}{4A_\Sigma q} - A_\Sigma q F, \\ &\dot{p}_N = -\mathcal{C}, \qquad \dot{p}_\eta = A_\Sigma N q\partial_\eta F, \quad \dot{p}_V = A_\Sigma N q\partial_V F. \end{aligned} \tag{H-01}$$

$$\partial_\eta F = -2\eta, \qquad \partial_V F = \frac{2q^2V}{9}, \qquad F_{(\eta,V)(\eta,V)} = \begin{pmatrix} -2 & 0 \\ 0 & 2q^2/9 \end{pmatrix}, \qquad \det = -\frac{4q^2}{9}. \tag{H-02}$$

For $q > 0$ the unique auxiliary solution is $\eta = V = 0$, with no condition on the NY coupling. The equivalent four constraints (p_η, p_V, η, V) have a nonsingular bracket matrix. The negative Hessian determinant describes an auxiliary quadratic saddle, not an instability of a propagating mode. EH, AX, VT and MX label restrictions of the input family; on the auxiliary shell they meet at the same torsionless configuration.

$$\{\phi_I, \phi_J\} = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \phi = (p_\eta, p_V, \eta, V), \qquad \det\{\phi_I, \phi_J\} = 1. \tag{H-03}$$

After eliminating the auxiliaries, the lapse and scale equations jointly give the following relations. We now use proper time τ.

$$\begin{aligned} F_* = \chi, \qquad & (Dq)^2 + \chi = 0, \\ 2qD^2q + (Dq)^2 + \chi = 0, \qquad & d\tau = N dt. \end{aligned} \tag{H-04}$$

Condition	Reduced trajectory	Interpretation
$\chi > 0$	No real trajectory	The real constraint surface is empty
$\chi = 0$	$q = q_0 > 0$	Constant positive scale
$\chi < 0$	$q = q_0 \pm \sqrt{-\chi}(\tau - \tau_0)$	Two directions, restricted to $q > 0$

Equation (H-04) gives $D^2q = 0$. For $\chi < 0$, the magnitude of Dq is a positive constant, so neither a bounce nor a recollapse occurs in a regular $q > 0$ interval. This follows analytically rather than from a numerical search. At $q = 0$ the nondegenerate-coframe assumption fails; no statement about full spacetime singularities or extensions is inferred from that endpoint.

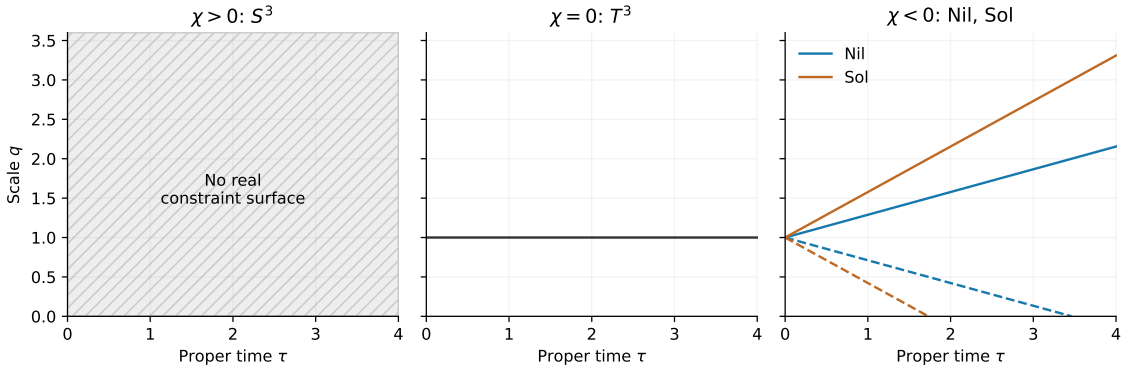


Figure 2: Reduced trajectories in proper time, with $q_0 = 1$ and only $q > 0$ displayed. Dashed curves contract and solid curves expand. Nil and Sol are stationary curves of the reduced action, but fail the omitted equations in Section 5.

For the torus, the reduced constraint $\mathcal{C} = -p_q^2/(4A_\Sigma q)$ has vanishing gradient on $p_q = 0$. This irregular presentation does not satisfy the hypothesis of the regular-constraint dimension formula. The Euler equations nevertheless determine the positive constant-scale solution directly.

Time-reparametrization consistency is verified by an off-shell Euler identity. The velocity Hessian of L_0, including all retained variables, has rank one: only the scale-velocity entry is nonzero. This is not a count of degrees of freedom in the full field theory.

$$
\begin{aligned}
E_z &= \partial L_0/\partial z - d(\partial L_0/\partial \dot{z})/dt, \\
\dot{q}E_q + \dot{\eta}E_\eta + \dot{V}E_V - N\frac{d}{dt}E_N &= 0, \\
\frac{\partial^2 L_0}{\partial \dot{z}\partial \dot{z}} &= \mathrm{diag}(-2A_\Sigma q/N, 0, 0, 0), \qquad z = (q, N, \eta, V).
\end{aligned}
\tag{H-05}
$$

4.1 Pure-scale directions on the torus

At a static solution with $\chi = 0$, $q_0 > 0$, put $q = q_0 + \epsilon u$ and select $N = 1$ only after lapse variation. The first and second coefficients of (V-01) are as follows, where $E^{[k]}$ denotes the coefficient of ϵ^k.

$$
\begin{aligned}
E_q^{[1]} &= 2A_\Sigma q_0 \ddot{u} = 0, \quad E_N^{[1]} = 0, \\
u &= a_u + b_u \tau, \qquad E_N^{[2]} = A_\Sigma q_0 b_u^2 = 0.
\end{aligned}
\tag{H-06}
$$

The time-linear direction of the local first-order equations is therefore excluded at second order. Fixed scale variations at both endpoints already set $a_u = b_u = 0$ at first order. On a fixed lattice, q_0 is a volume modulus, not a time-reparametrization gauge. The proper duration $T = \int N dt$ is invariant under endpoint-fixing time transformations; at fixed T, lapse variations of zero integral are gauge. The invertible coordinate and endpoint-representative changes in Appendix C preserve the irregular constraint. Section 5 allows shape directions.

5 Common-scale reduction and shape evolution

Complete NY with constant coupling does not change the bulk connection equation. Independent variation of the metric-compatible connection in the EC action, with no curvature-square term or spin source, gives the Cartan equation below. For a nondegenerate coframe it sets all torsion to zero. Appendix C proves this without imposing the two-amplitude ansatz.

$$
\epsilon_{abcd}\, e^c \wedge T^d = 0 \quad \Longrightarrow \quad T^a = 0. \tag{E-01}
$$

After torsion is eliminated, the full coframe equation is the vacuum Einstein equation. On the common-scale metric it splits as follows. The matrix Ξ is the spatial tracefree Ricci tensor and is determined by the structure matrix.

$$
\begin{aligned}
G_{00} &= 3(H^2 + \chi/q^2), \qquad G_{0i} = 0, \\
G_{ij} &= (-2DH - 3H^2 - \chi/q^2)\delta_{ij} + \Xi_{ij}, \\
\Xi_{ij} &= {}^{(3)}\mathrm{Ric}_{ij} - \frac{2\chi}{q^2}\delta_{ij}.
\end{aligned}
\tag{E-02}
$$

Imposing (H-04) removes G_{00} and the spatial trace, but leaves Ξ. In diagonal frames the four residuals are

$$
q^2\Xi_{T^3} = q^2\Xi_{S^3} = 0, \qquad q^2\Xi_{\mathrm{Nil}} = \tfrac{1}{3}\mathrm{diag}(-1, -1, 2), \qquad q^2\Xi_{\mathrm{Sol}} = \tfrac{1}{3}\mathrm{diag}(-4, 2, 2). \tag{E-03}
$$

Consequently the constant-scale torus satisfies the full vacuum equations, whereas the sphere has no real trajectory even in the reduction. The linear Nil and Sol trajectories are stationary curves of the fixed-shape variational problem, not full vacuum solutions. This does not exclude vacuum solutions with time-dependent anisotropic shapes on Nil or Sol.

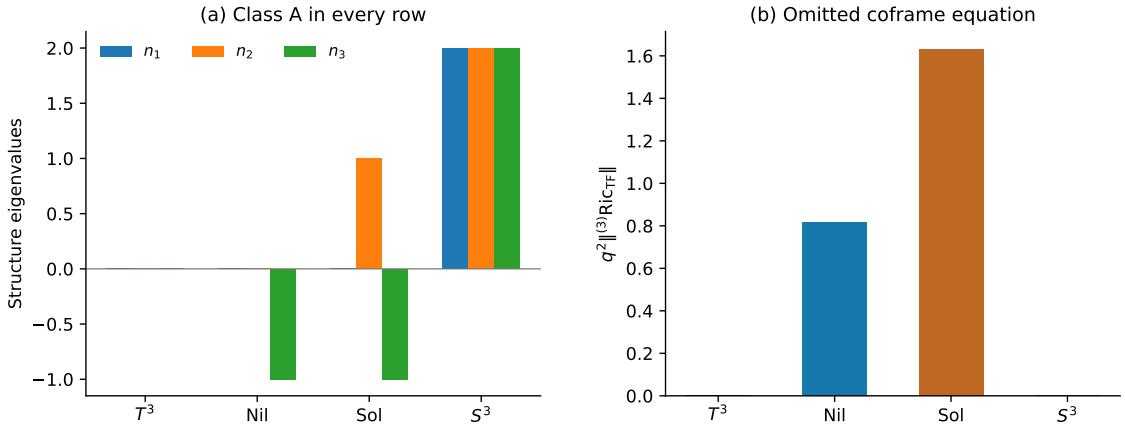


Figure 3: All four representatives are class A. Left: structure eigenvalues. Right: the Euclidean norm of spatial tracefree Ricci. Where a real reduced trajectory exists, this is its remaining Einstein residual; S^3 has no such trajectory. Its unit-scale norm is $\sqrt{J_2}$; the negative unit-scale tensor gives the initial shape acceleration in (S-03). The norm is neither a physical mass nor an instability exponent.

For this action, the independent connection equation fixes all torsion, and the full coframe equation supplies the spatial-curvature condition remaining on a reduced trajectory. Algebraic auxiliary elimination, existence of real constraint data and satisfaction of the full field equations can thus be examined in sequence.

5.1 Condition for a common-scale vacuum family

$$\begin{aligned} f(n) &= \tfrac{1}{2}(\mathrm{tr}n)^2 - \mathrm{tr}(n^2) = 6\chi, \\ X(n) &= q^2\Xi = 2n^2 - (\mathrm{tr}n)n + 4\chi I, \qquad \mathrm{tr}X = 0, \\ \delta L_{\mathrm{shape}} &= -\frac{N\mathcal{V}_\Sigma q}{\kappa^2}\mathrm{tr}(X\delta\mathsf{B}). \end{aligned} \tag{E-04}$$

$$X = 0 \quad \Longleftrightarrow \quad n \sim \mathrm{diag}(a,a,a) \text{ or } \mathrm{diag}(a,a,0), \qquad a \in \mathbb{R}. \tag{E-05}$$

Here $\sim$ denotes constant $SO(3)$ conjugacy. The first branch has $\chi = a^2/4$, so nonzero a admits no real common-scale vacuum trajectory. The second is spatially flat with $\chi = 0$ and gives static real vacuum families. This is a local classification for real symmetric n, not an existence assertion for new compact quotients. Appendix D proves it by five-direction variation and eigenvalue differences.

5.2 Mixed constraint cone of the flat torus

On the fixed-lattice $n = 0$ torus, set $q = q_0 e^{\epsilon\rho}$, $\mathsf{B} = \epsilon\mathsf{B}_1$, $N = N_0 + \epsilon\delta N$ and $\tau = \int N_0 dt$. The quadratic action, its linear solutions and the second-order coefficient of the original lapse equation give

$$\begin{aligned} L^{[2]} &= \frac{A_\Sigma q_0^3}{6N_0}\left[\mathrm{tr}(\dot{\mathsf{B}}_1^2) - 6\dot{\rho}^2\right], \\ \rho &= \rho_0 + h\tau, \qquad \mathsf{B}_1 = \mathsf{B}_0 + \Sigma\tau, \\ \Sigma^T &= \Sigma, \quad \mathrm{tr}\Sigma = 0, \qquad \mathrm{tr}(\Sigma^2) = 6h^2. \end{aligned} \tag{S-01}$$

If $\Sigma = 0$, then $h = 0$, in agreement with (H-06) through $u = q_0\rho$. Nonzero mixed directions are integrable. For each direction with $h \neq 0$, define the following exact metric on the fixed lattice; $\psi^{2\mathsf{P}} = \exp(2\mathsf{P}\log\psi)$ is positive definite.

$$\begin{aligned} \mathsf{P} &= \tfrac{1}{3}I + \frac{\Sigma}{3h}, \quad M_0 = \rho_0 I + \mathsf{B}_0, \quad \psi = 1 + 3\epsilon h\tau > 0, \\ g_\epsilon(\tau) &= q_0^2 e^{\epsilon M_0}\psi^{2\mathsf{P}} e^{\epsilon M_0}, \\ \mathrm{tr}\mathsf{P} &= \mathrm{tr}(\mathsf{P}^2) = 1, \qquad q_\epsilon = q_0 e^{\epsilon\rho_0}\psi^{1/3}. \end{aligned} \tag{S-02}$$

This is a Kasner-type vacuum family [12], with tangent ρ, B_1 at $\epsilon = 0$. On any bounded time interval, $\psi > 0$ holds uniformly for sufficiently small ϵ. When $h = 0$, the cone forces $\Sigma = 0$ and static positive definite metric moduli realize the direction. In this homogeneous sector, the cone is thus necessary and sufficient for local integrability of each first-order tangent. This is the homogeneous form of the linearization instability of the flat torus [14]: the cone is the second-order condition attached to the static time translation, and the momentum constraints vanish identically for $n = 0$ (Appendix D). Fixing the metric at both endpoints instead sets all affine first variations to zero. Appendix D includes noncommuting initial shape.

5.3 Initial shape evolution on Nil and Sol

In the diagonal LC vacuum sector choose $q = q_0$, $\beta_i = D\beta_i = 0$ and $H_0^2 = -\chi/q_0^2$ initially, with $e^i = qe^{\beta_i}\sigma^i$, $\sum_i \beta_i = 0$. Set $\hat\tau = \tau/q_0$ and $h_* = H_0 q_0$. The initial X_i and $J_2 = \mathrm{tr}(X^2)$ determine the exact Taylor coefficients

$$\begin{aligned} \beta_i &= -\tfrac{1}{2}X_i\hat\tau^2 + \tfrac{5}{6}h_* X_i\hat\tau^3 + O(\hat\tau^4), \\ q/q_0 &= 1 + h_*\hat\tau - \tfrac{1}{36}J_2\hat\tau^4 + O(\hat\tau^5). \end{aligned} \tag{S-03}$$

Geometry	J_2	Coefficient of $\hat\tau^4$ in q/q_0
Nil	$2/3$	$-1/54$
Sol	$8/3$	$-2/27$

The residual Ξ drives the quadratic shape evolution, while J_2 fixes the quartic departure from linear scale. Appendix D derives the closed ODE, constraint propagation and coefficients. The supplement compares 12 numerical points across both time orientations; these check the local coefficients and do not certify an error bound over an interval or long-time stability.

6 Time-dependent curvature and relative endpoint quantities

Compute the curvature of the same connection (G-04) before imposing static restrictions. The three spatial planes have distinct y_i, while the other blocks x, z, w are common.

$$\begin{aligned} R^0{}_i &= xe^{0i} + we^{jk}, \qquad R^j{}_k = ze^{0i} + y_i e^{jk}, \\ x &= Db + Hb, \qquad z = -D\eta/q, \qquad w = 2b\eta/q, \\ y_i &= b^2 + (k_i - \eta^2)/q^2, \qquad \bar y = \tfrac{1}{3}\sum_i y_i = b^2 + (\chi - \eta^2)/q^2. \end{aligned} \tag{D-01}$$

Define Q, complete NY and Pontryagin as oriented forms. In the internal-dual contraction $\mathcal{B}$, the form indices are raised with the Lorentz metric. With all internal indices summed, direct wedge products and component contractions give

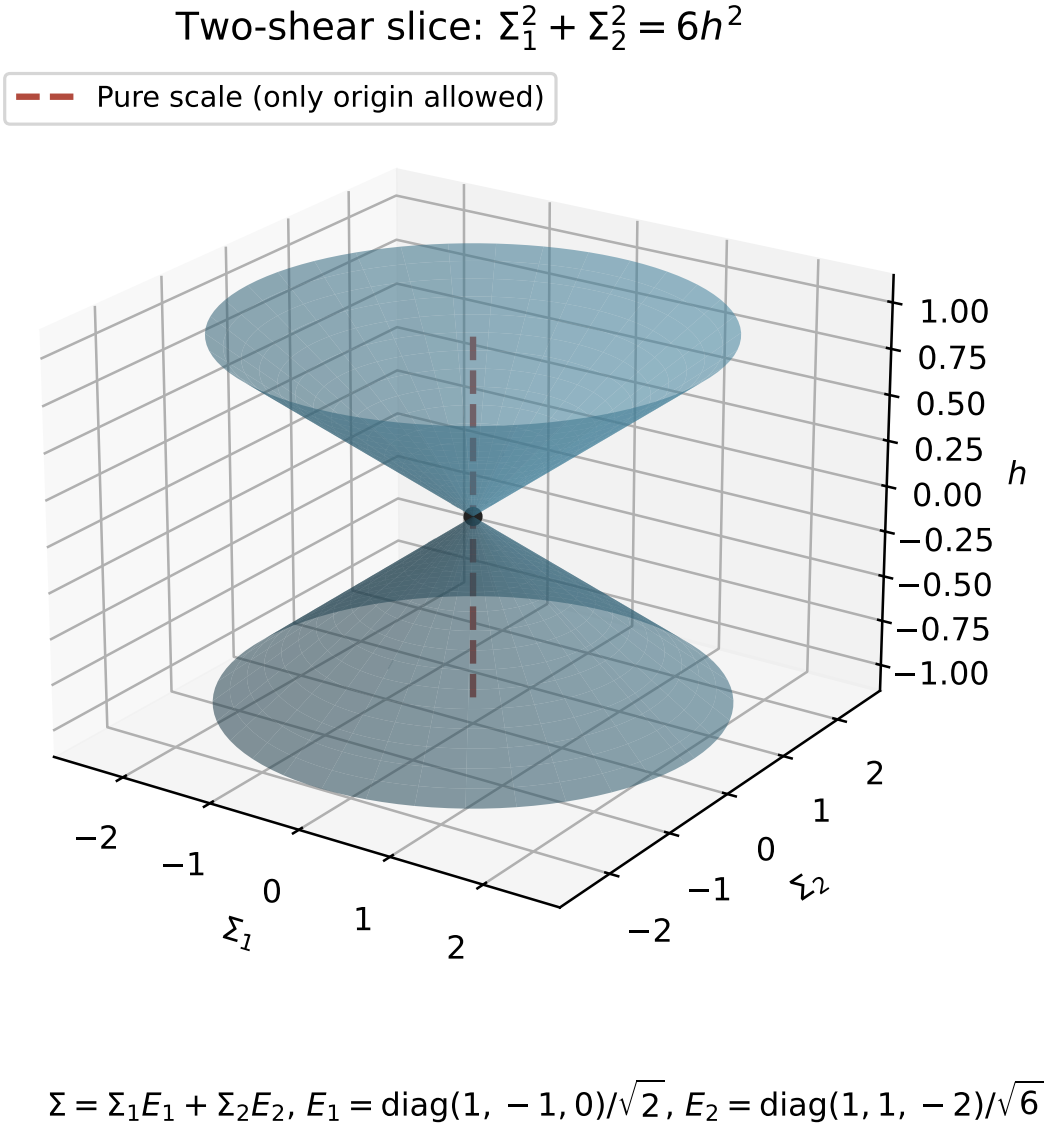


Figure 4: Three-dimensional slice of $\mathrm{tr}(\Sigma^2) = 6h^2$ using two orthonormal shear directions. The pure-scale axis meets the cone only at the origin. This is neither the full five-shape cone nor a physical degree-of-freedom space.

$$
\begin{aligned}
Q &= \frac{6\eta}{q} e^{123}, & Q_\Sigma &= 6\mathcal{V}_\Sigma q^2 \eta, \\
dQ &= \mathcal{N} e^{0123}, & \mathcal{N} &= 6(D\eta + 2H\eta)/q, \\
\sum_{a,b} R_{ab} \wedge R^{ab} &= \mathcal{P} e^{0123}, & \mathcal{P} &= 12(\bar{y}z - xw), \\
\mathcal{B} &= \tfrac{1}{16} \epsilon^{abef} R_{abcd} R_{ef}{}^{cd} = 3(w\bar{y} - xz).
\end{aligned}
\tag{D-02}
$$

Summing the spatial-plane contributions gives $\sum_i y_i = 3\bar{y}$, so the structure dependence of these scalar expressions reduces to χ. This identity uses a constant class-A structure matrix, a common scale and the two-amplitude torsion ansatz. It is not an identity for arbitrary fields with time-dependent shape or tensor torsion. The duals in $\mathcal{B}$ and $\mathcal{P}$ act on different index pairs, and the two quantities are distinct.

Family	Local quantities	Endpoint quantity
Static	$\mathcal{N} = \mathcal{P} = 0$	Q_Σ can be nonzero
Dynamic AX	$\mathcal{N}, \mathcal{P}, \mathcal{B}$ need not vanish	Q_Σ can be nonzero
Dynamic VT	$\mathcal{N} = \mathcal{P} = \mathcal{B} = 0$	$Q_\Sigma = 0$
Dynamic MX	Given by (D-02)	Q_Σ can be nonzero
Smooth auxiliary shell	$\mathcal{N} = \mathcal{P} = \mathcal{B} = 0$	$Q_\Sigma = 0$

Static means that all time derivatives of q, η, V vanish. For dynamic VT, η vanishes identically throughout the interval, so $D\eta = 0$ as well. The auxiliary shell likewise imposes $\eta = V = 0$ identically. A profile whose amplitudes vanish at only one instant is not the same restriction.

$$
\mathcal{B}|_{\text{static}} = \frac{2V\eta}{9q^3}(-V^2q^2 + 9\eta^2 + C), \qquad C = -9\chi. \tag{D-03}
$$

Equation (D-03) is the static limit of the covariant internal-dual contraction. Static AX and VT give zero, while static MX can be nonzero. Extracting the $V\eta$ coefficient of the curvature contraction determines C, as derived in Appendix B.

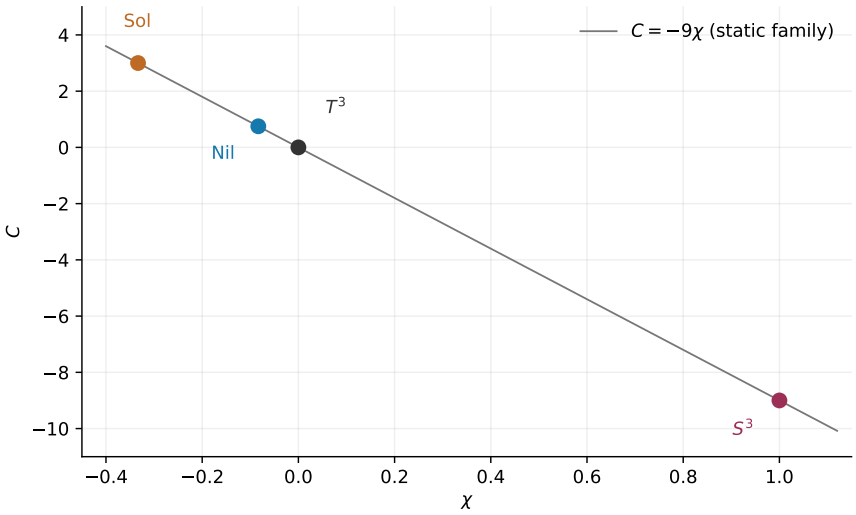


Figure 5: The coefficient C extracted from the covariant static internal-dual contraction versus χ derived from LC spatial curvature. The physical-radius sphere has $\chi = 1$ and $C = -9$.

Time dependence also contributes to the Pontryagin density. The following jet, realizable by a smooth off-shell profile, gives $\mathcal{P} = -9$ on the sphere.

$$q = N = 1, \quad \dot{q} = \ddot{q} = \dot{N} = V = \dot{V} = 0, \quad \eta = \tfrac{1}{2}, \quad D\eta = 1 \quad \Longrightarrow \quad \mathcal{P} = 3 - 12\chi, \quad \mathcal{N} = 6. \tag{D-04}$$

6.1 Absolute and relative primitives

Use the local CS form of the same connection. Let s_1, s_2, s_3 be the elementary symmetric polynomials in the diagonal structure eigenvalues. Direct calculation gives the slice primitive

$$\begin{aligned} \mathrm{CS}_{\mathcal{P}} &= -\mathrm{tr}(\omega \wedge d\omega + \tfrac{2}{3}\omega \wedge \omega \wedge \omega), \\ \int_\Sigma \mathrm{CS}_{\mathcal{P}} &= \mathcal{V}_\Sigma(\Phi_0 + \phi), \\ \Phi_0 &= s_1^3 - 4s_1 s_2 + 10 s_3, \qquad s_1 = \sum_i n_i, \quad s_2 = \sum_{i<j} n_i n_j, \quad s_3 = n_1 n_2 n_3, \\ \phi &= -12\chi\eta + 4\eta^3 - 12\eta q^2 b^2, \qquad D\phi = q^3 \mathcal{P}. \end{aligned} \tag{D-05}$$

The constant Φ_0 is the LC reference value in the specified homogeneous frame. It equals $0, -1, 0, 8$ for T^3, Nil, Sol and S^3, respectively. It drops out of endpoint differences on a fixed-structure time interval, but is not determined by χ alone. Absolute CS values in changing local framings are not assigned the meaning of invariant charges.

Choose the LC connection $\bar{\omega}$ of the same coframe as reference and set $K = \omega - \bar{\omega}$. With both connections compared in the same frame identification, define the relative form

$$\begin{aligned} \mathcal{T}_{\mathcal{P}} &= -2\mathrm{tr}(K \wedge \bar{R}) - \mathrm{tr}(K \wedge \bar{D}K) - \tfrac{2}{3}\mathrm{tr}(K \wedge K \wedge K), \\ d\mathcal{T}_{\mathcal{P}} &= -\mathrm{tr}(R \wedge R) + \mathrm{tr}(\bar{R} \wedge \bar{R}), \\ \int_\Sigma \mathcal{T}_{\mathcal{P}} &= \mathcal{V}_\Sigma \phi, \qquad \int_M \mathcal{P} e^{0123} = \mathcal{V}_\Sigma [\phi]_i^f. \end{aligned} \tag{D-06}$$

LC Pontryagin vanishes on this family, so the relative primitive differentiates to $\mathcal{P}$. Under a simultaneous local Lorentz transformation, $K, \bar{R}, \bar{D}K$ transform by conjugation; cyclicity of the trace makes the relative form invariant. This does not compare independently chosen endpoint frames without an identification or assert quantization. The NY integral likewise equals the oriented difference of Q_Σ.

6.2 Selected quadratic and cubic contractions

This subsection concerns the common-scale, two-amplitude family. Use $x = Db + Hb$, $\bar{y} = b^2 + (\chi - \eta^2)/q^2$, $z = -D\eta/q$, $w = 2b\eta/q$. All indices are raised and lowered with the Lorentz metric; pair-exchange symmetry of EC curvature is not assumed.

$$\begin{aligned} J_2 &= \mathrm{tr}(X^2), & J_3 &= \mathrm{tr}(X^3), \\ I_1 &= R_{abcd} R^{abcd}, & I_2 &= R_{abcd} R^{cdab}, \\ I_{\mathrm{Ric}} &= \mathrm{Ric}_{ab} \mathrm{Ric}^{ab}, & C_{\mathrm{Ric}} &= \mathrm{Ric}^a{}_b \mathrm{Ric}^b{}_c \mathrm{Ric}^c{}_a, \\ C_R &= R_{ab}{}^{cd} R_{cd}{}^{ef} R_{ef}{}^{ab}. \end{aligned} \tag{I-01}$$

C_R sums all ordered indices, without the factors of $1/2$ sometimes used for curvature operators on two-forms. Direct contraction of the curvature blocks gives

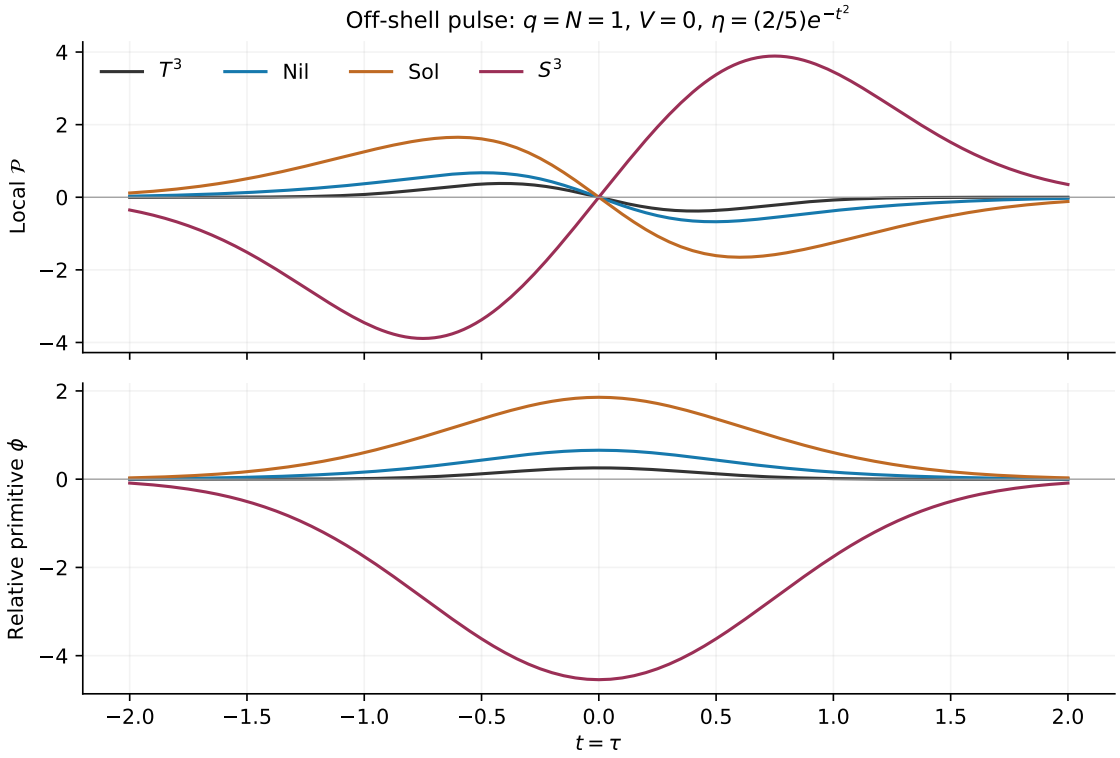


Figure 6: An off-shell pulse at fixed scale. Local $\mathcal{P}$ is generally nonzero, but its interval integral vanishes because the even profile has equal relative primitives at the endpoints. Local densities and endpoint quantities are not identified with physical response.

$$\begin{aligned} I_1 &= 12(x^2 + \bar{y}^2 - w^2 - z^2) + 4J_2/q^4, \\ I_2 &= 12(x^2 + \bar{y}^2 + 2wz) + 4J_2/q^4, \\ I_{\mathrm{Ric}} &= 12(x^2 + x\bar{y} + \bar{y}^2) + J_2/q^4, \\ I_1 - I_2 &= -12(w + z)^2. \end{aligned} \tag{I-02}$$

$$\begin{aligned} C_{\mathrm{Ric}} &= 27x^3 + 3(x + 2\bar{y})^3 + 3(x + 2\bar{y})J_2/q^4 + J_3/q^6, \\ C_R &= 24[x^3 + \bar{y}^3 + 3wz(x + \bar{y})] + 24\bar{y}J_2/q^4 - 8J_3/q^6. \end{aligned} \tag{I-03}$$

$$C_{abcd}C^{abcd}\Big|_{\mathrm{LC}} = 2J_2/q^4. \tag{I-04}$$

Here C_{abcd} is the LC Weyl tensor; no EC Weyl interpretation is implied. On the linear reduced Nil and Sol trajectories, $x = \bar{y} = w = z = 0$ still leaves $I_{\mathrm{Ric}} = J_2/q^4 \neq 0$. Vanishing selected reduced scalars therefore does not imply vanishing curvature. Appendix B derives the contractions and trace subtraction.

7 Curvature information and geometric representations

Equations (B-03), (D-02) and (D-05) organize the reduced bulk action, local NY, Pontryagin and internal-dual scalars, and the relative primitive per reference volume by χ. The curvature tensor itself retains the three separate k_i. Scalar-curvature control therefore does not identify the geometries or their full field equations.

For an explicit equal-χ comparison, take Nil with $n = \mathrm{diag}(0, 0, -1)$ and a Sol-type reference with $n = \mathrm{diag}(0, 1/2, -1/2)$. The latter uses a different reference normalization from the canonical Sol row. Both have $\chi = -1/12$, but

$$\begin{aligned} q^2\Xi_{\mathrm{Nil}} &= \mathrm{diag}(-1/3, -1/3, 2/3), \qquad \Phi_{0,\mathrm{Nil}} = -1, \\ q^2\Xi_{\mathrm{Sol}/2} &= \mathrm{diag}(-1/3, 1/6, 1/6), \qquad \Phi_{0,\mathrm{Sol}/2} = 0. \end{aligned} \tag{U-01}$$

This is a comparison of local structure matrices, not an identification of quotient volumes or lattices. Integrated actions and endpoint quantities always carry $\mathcal{V}_\Sigma$. The sign of χ distinguishes reduced trajectories; it does not establish topology selection by comparing absolute action values.

The connection with the compact geometries of Paper04 is expressed through structure matrices and closed one-forms. Their ranks are $0, 1, 2, 3$, and the dimensions of homogeneous closed one-forms are $3, 2, 1, 0$ for T^3, Nil, Sol and S^3. Paper04 treats harmonic one-forms and quadratic forms on the specified compact quotients. These are geometric data characterizing the fixed coframes used here.

Bianchi class is determined by the trace of the structure constants. Appendix A evaluates the Jacobi condition and trace in a five-parameter representation and identifies its class-A sector, including Sol.

7.1 Restricted χ-universality and the information hierarchy

Here restricted χ-universality means precisely that, in the specified common-scale, two-amplitude family, the bulk action, local NY, Pontryagin and internal-dual scalars, and relative primitive per reference volume depend on n through χ. It does not assert universality of arbitrary curvature contractions or shape equations.

Quantity	Structure information required
Reduced action and (D-02), (D-05)	χ
Selected quadratic contractions, LC Weyl square	χ, J_2
Selected cubic contractions	χ, J_2, J_3
Absolute CS, integrated quantities	Frame, structure polynomial and quotient volume also enter

The equal-χ examples in (U-01) have $J_2 = 2/3$ and $1/6$. The following pair has equal χ, J_2 but distinct J_3. It compares local off-shell curvatures at positive χ, not real common-scale vacuum solutions.

n	X	(χ, J_2, J_3)
$(4, 4, 1)$	$(1, 1, -2)$	$(5/4, 6, -6)$
$(2, 2, 3)$	$(-1, -1, 2)$	$(5/4, 6, 6)$

At the same time variables, first minus second gives $\Delta C_{\mathrm{Ric}} = -12/q^6$, $\Delta C_R = 96/q^6$. The tracefree Ricci spectrum is fixed by the polynomial below; adding 2χ to each root gives the unit-scale spatial Ricci spectrum.

$$\det(\lambda I - X) = \lambda^3 - \tfrac{1}{2}J_2\lambda - \tfrac{1}{3}J_3, \qquad J_2 \geq 0, \quad \tfrac{1}{2}J_2^3 - 3J_3^2 \geq 0. \tag{I-05}$$

This reconstructs the Ricci spectrum at a point. For the inverse problem for n, three nonzero Ricci eigenvalues are realizable precisely when their product is positive; the solution is unique up to conjugacy and overall sign. Exactly one nonzero eigenvalue allows continuous nonuniqueness; exactly two is impossible. Zero Ricci gives the flat branch. Appendix B proves these statements. Nonunique structure matrices are not asserted to define locally nonisometric metrics.

7.2 Frames on the same flat torus

On the same 2π-periodic T^3, rotate the coordinate coframe through the angle νx^3, $\nu \in \mathbb{Z}$. The resulting $n = (-\nu, -\nu, 0)$ represents the same static flat metric as $n = 0$. Its connection is pure gauge and its vector holonomy is the identity around every period. Appendix A gives the coframe and period calculation.

Frame	rank n	Closed constant-coefficient 1-forms	$b_1(T^3)$
Coordinate	0	3	3
Rotated, $\nu \neq 0$	2	1	3

The number of closed forms within a fixed constant-coefficient coframe sector must therefore be distinguished from de Rham cohomology. This clarifies representation dependence while retaining the results for the specified coframes and quotients of Paper04. No spin-holonomy statement is made.

8 Conclusions

For Lorentzian EC gravity with constant complete NY coupling and no matter, cosmological or curvature-square term, we compared four fixed reference geometries. The common-scale torsion amplitudes are auxiliary and real reduced trajectories are organized by the sign of χ. Combining the full connection and shape equations restricts real common-scale vacuum families to locally flat static ones.

With shape allowed, the second-order obstruction to pure scale on the flat torus becomes part of a mixed scale–shear cone. Each tangent on this cone is realized by an exact local vacuum family on the fixed lattice. For Nil and Sol, the initial tracefree Ricci tensor fixes quadratic shape evolution and J_2 fixes the quartic scale departure.

Restricted χ dependence holds for the specified common-scale, two-amplitude scalars and relative primitive. Selected quadratic contractions additionally require J_2, and cubic contractions require J_3. These three quantities recover the spatial Ricci spectrum at a point, but degenerate branches of the inverse structure-matrix problem are nonunique. The flat-torus frame comparison and canonical endpoint correspondence make the distinctions between structure representations, cohomology and boundary data explicit.

The arguments are contained in the main text and Appendices A–D; the supplement provides rederivation, saved-data rendering and numerical comparisons at specified points. General inhomogeneous fields, long-time anisotropic evolution, physical spectra or stability, classification of compact quotients, covariant-derivative invariants and responses to a dynamical NY coupling are outside the present claims.

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This work was developed within the informal collaborative project **DPPU** (**D**onut-like topology, **P**lanck-scale compactness, **P**recession dynamics, **U**niverse).

Availability and contact

The manuscript, code and data are available in https://github.com/Muacca/DPPUv7-paper06. The existing Zenodo record is doi:10.5281/zenodo.20228881; cite the edition and date of the artifact actually obtained. The supplement is released under the MIT License. Contact: muacca@dmwp.jp.

A Structure constants and component conventions

A.1 Jacobi identity and Bianchi class

Define the structure constants by $d\sigma^i = \frac{1}{2}C^i{}_{jk}\sigma^j \wedge \sigma^k$ and represent them by five coefficients. Their nonzero scale-stripped entries are listed below, with antisymmetric lower indices. The Jacobi residual and trace give the conditions for this representation to define a Lie algebra and to be unimodular.

$$\begin{aligned} &C^1{}_{23} = a, \quad C^2{}_{31} = b, \quad C^3{}_{12} = c, \quad C^2{}_{12} = u, \quad C^3{}_{13} = v, \\ &J_{123} = (-a(u+v), 0, 0), \qquad C^j{}_{ij} = (u+v, 0, 0). \end{aligned} \tag{A-01}$$

Thus the Lie family requires $a(u+v) = 0$, and the unimodular sector used in the paper has $u + v = 0$; it is (G-01) with $n = \mathrm{diag}(a, b, c)$ and $n_{23} = n_{32} = u$. Polynomial identities outside the Jacobi locus are not statements about Lie-group curvature. Sol with $u = 1, v = -1$ is class A; an orientation-preserving rotation by $\pi/4$ in the 23-plane diagonalizes its structure matrix. The general formulas in the text refer to this class-A sector.

A.2 Component convention for complete NY

With the oriented alternating symbol $[0123] = +1$, the component coefficient in complete NY is $1/4$ on the torsion side and $1/2$ on the curvature side. This symbol is distinct from the raised Lorentz epsilon tensor.

$$\mathcal{N} = \tfrac{1}{4}[abcd]T^e{}_{ab}T_{ecd} - \tfrac{1}{2}[abcd]R_{abcd}. \tag{A-02}$$

A.3 Periodic rotating coframe on a fixed torus

Let x^1, x^2, x^3 have period 2π and $\nu \in \mathbb{Z}$. The following $O(x^3) \in SO(3)$ is global, and $\sigma = O(dx^1, dx^2, dx^3)^T$ is an orthonormal coframe for the same flat metric.

$$\begin{aligned} O(x^3) &= \begin{pmatrix} \cos\nu x^3 & \sin\nu x^3 & 0 \\ -\sin\nu x^3 & \cos\nu x^3 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ d\sigma^1 &= -\nu\sigma^2 \wedge \sigma^3, \quad d\sigma^2 = -\nu\sigma^3 \wedge \sigma^1, \quad d\sigma^3 = 0, \\ n &= \mathrm{diag}(-\nu, -\nu, 0), \qquad \omega = -dO\,O^T, \quad \omega^1{}_2 = -\nu dx^3. \end{aligned} \tag{F-01}$$

Direct calculation gives $d\sigma + \omega \wedge \sigma = 0$ and $d\omega + \omega \wedge \omega = 0$. Integer periodicity gives $O(x^3 + 2\pi) = O(x^3)$, so vector parallel transport is the identity around each closed period. For $\nu \neq 0$, constant

α_i satisfy $d(\alpha_i\sigma^i) = 0$ only when $\alpha_1 = \alpha_2 = 0$. In contrast, dx^1, dx^2, dx^3 are global closed forms with variable coefficients in this coframe; their period matrix on the three coordinate cycles is $2\pi I$.

The upper bound $b_1 = 3$ also follows from Fourier decomposition on the flat torus: each nonzero-wavevector component of a closed one-form is parallel to its wavevector and is exact. Only three zero-mode components remain in cohomology. Thus the number of closed forms in the constant-coefficient coframe subspace is not a Betti number. This comparison concerns the specified static metric and vector connection, not spin structures or equivalence of the full dynamical ansatz in both frames.

B Derivation of the curvature identities

In a diagonal frame, $de^i = He^{0i} + (n_i/q)e^{jk}$. The spatial coefficients in (G-04) obey the following relation, so the spatial torsion from the first Cartan equation is $2\eta/q$.

$$a_j + a_k = \frac{n_i - 2\eta}{q}, \qquad \frac{n_i}{q} - a_j - a_k = \frac{2\eta}{q}, \qquad H - b = V/3. \tag{R-01}$$

Expanding the second Cartan equation gives temporal boost coefficient $Db + Hb$ and spatial boost coefficient $b(n_i/q - a_j - a_k)$. The rotation coefficients are given below. Cancellation of the spatial term linear in η is the key to the common formula.

$$\begin{aligned} Da_i + Ha_i &= -D\eta/q, \\ b^2 + \frac{n_i a_i}{q} - a_j a_k &= b^2 + \frac{n_i\ell_i - \ell_j\ell_k - \eta^2}{q^2}, \\ \ell_j + \ell_k &= n_i. \end{aligned} \tag{R-02}$$

This proves (D-01). Spatial scalar and Ricci curvature follow by setting torsion, b and Db to zero. Summing the k_i yields (G-02) and the four representatives give the displayed χ and tracefree Ricci tensors. Every real symmetric n can be brought to this frame by a constant $SO(3)$ rotation, so the scalar identities are not artifacts of a preferred diagonal display.

In the Pontryagin wedge summed over all internal indices, the three boost pairs contribute $-12xw$ and the rotation pairs contribute $4z\sum_i y_i$. The internal-dual contraction weights purely spatial form pairs by $+1$ and temporal form pairs by -1, giving $w\sum_i y_i - 3xz$. These contractions prove (D-02).

On the static restriction $H = Db = D\eta = 0$, $b = -V/3$, one obtains $\mathcal{B} = 3w\bar{y}$. Extracting the coefficient linear in $V\eta$ from this polynomial yields $C = -9\chi$.

NY is also checked by direct wedge products. The ηV terms cancel between the two sides, leaving precisely the time derivative of Q.

$$\begin{aligned} T^a \wedge T_a &= \frac{4V\eta}{q} e^{0123}, \\ e^a \wedge e^b \wedge R_{ab} &= \left(\frac{4V\eta}{q} - \frac{6(D\eta + 2H\eta)}{q}\right) e^{0123}. \end{aligned} \tag{R-03}$$

Expansion of the CS form from the connection gives the following constant part at $\eta = 0$. Expressing it in elementary symmetric polynomials yields Φ_0 in (D-05).

$$\Phi_0 = \sum_i n_i^3 - \sum_{i\neq j} n_i^2 n_j + 4n_1n_2n_3 = s_1^3 - 4s_1s_2 + 10s_3. \tag{R-04}$$

Differentiating the remaining ϕ and using $Dq = Hq$, $Db = x - Hb$ gives $-12(\chi - \eta^2)D\eta - 12q^2b^2D\eta - 24q^2b\eta x = q^3\mathcal{P}$. This is also an independent comparison between the local density and the endpoint primitive.

The general relative form (D-06) follows from the connection path $\omega_\gamma = \bar{\omega} + \gamma K$, with $R_\gamma = \bar{R} + \gamma \bar{D}K + \gamma^2 K \wedge K$. Bianchi identity and cyclicity give the formula below; integration from $\gamma = 0$ to 1 proves (D-06). This argument does not depend on the four representatives.

$$\partial_\gamma[-\mathrm{tr}(R_\gamma \wedge R_\gamma)] = -2d\,\mathrm{tr}(K \wedge R_\gamma). \tag{R-05}$$

B.1 Ordered contractions and the LC Weyl square

If X_i are the eigenvalues of X, then $y_i = \bar{y} - X_i/q^2$ and $\sum_i X_i = 0$. Hence $\sum y_i = 3\bar{y}$, $\sum y_i^2 = 3\bar{y}^2 + J_2/q^4$, and $\sum y_i^3 = 3\bar{y}^3 + 3\bar{y}J_2/q^4 - J_3/q^6$. Ordered-index contraction of the curvature blocks gives

$$\begin{aligned} I_1 &= 4\sum_i (x^2 + y_i^2 - w^2 - z^2), \\ I_2 &= 4\sum_i (x^2 + y_i^2 + 2wz), \\ C_R &= 8\sum_i [x^3 + y_i^3 + 3wz(x + y_i)], \\ \mathrm{Ric}^a{}_b &= \mathrm{diag}(3x, x + 2\bar{y} + X_1/q^2, x + 2\bar{y} + X_2/q^2, x + 2\bar{y} + X_3/q^2). \end{aligned} \tag{I-06}$$

The differing negative and cross terms in the first two lines follow from the Lorentz metric and the exchanged index pairs. Squaring and cubing the last diagonal matrix gives the Ricci contractions in (I-02), (I-03). In the LC restriction, $w = z = 0$ and $R = 6(x + \bar{y})$. Substitution into the four-dimensional trace subtraction $C^2 = I_1 - 2I_{\mathrm{Ric}} + R^2/3$ cancels all $x, \bar{y}$ terms and proves (I-04). These calculations derive the selected contractions without an external code prerequisite.

B.2 Ricci spectrum and inverse structure matrix

For real symmetric tracefree X, Newton identities give $\sum_{i<j} X_i X_j = -J_2/2$ and $\det X = J_3/3$, proving (I-05). Three real roots require nonnegative discriminant $J_2^3/2 - 3J_3^2$; conversely this condition with $J_2 \geq 0$ gives three real roots. The case $J_2 = 0$ means $X = 0$, while zero discriminant means repeated roots. Neither is to be conflated with a zero Ricci eigenvalue below.

At unit scale, let $r_i = X_i + 2\chi$ be the spatial Ricci eigenvalues. With $s_1 = \mathrm{tr}n$ and $\ell_i = s_1/2 - n_i$, one has $n_i = \ell_j + \ell_k$ and $r_i = 2\ell_j\ell_k$, directly by substitution into $k_j + k_k$. At physical scale q, the eigenvalues are r_i/q^2.

$$r_i \neq 0\ (\forall i): \qquad r_1 r_2 r_3 > 0, \qquad \ell_i^2 = \frac{r_j r_k}{2r_i}, \qquad n_i = \ell_j + \ell_k. \tag{I-07}$$

In the nondegenerate branch, positive product is necessary and also sufficient: the magnitudes in (I-07) and the sign conditions $r_i = 2\ell_j\ell_k$ construct the solution. Choosing the sign of ℓ_1 fixes $\ell_2 = r_3/(2\ell_1)$ and $\ell_3 = r_2/(2\ell_1)$; the remaining equation follows automatically. Thus only the overall sign and orthogonal eigenspace basis remain. Permutations and conjugacies of real symmetric matrices can be implemented by $SO(3)$ conjugacy.

If two r_i are nonzero, all three ℓ_i are nonzero, forcing the third r_i to be nonzero: exactly two is impossible. If $r_1 \neq 0$ and $r_2 = r_3 = 0$, then $\ell_1 = 0$, $\ell_2\ell_3 = r_1/2$, leaving a continuous product family. For example, $\ell = (0, t, t^{-1})$, $t \neq 0$, gives $n = (t + t^{-1}, t^{-1}, t)$ with the same $r = (2, 0, 0)$.

If all $r_i = 0$, at most one ℓ_i is nonzero, so $n \sim (a, a, 0)$. This is an algebraic inverse problem for structure matrices, not a classification of local isometry or compact quotients.

C Variations and the full connection equation

Substitution of (B-02) into the raw scalar density and integration by parts of the EH acceleration and vector divergence give (B-03). The vector terms after spatial integration are $-\mathcal{V}_\Sigma d(q^3 V)/\kappa^2$. The HV term alone must not be discarded as a total derivative.

The Euler derivatives of L_0 are listed below. In $\partial_q F$, the amplitudes η, V are held fixed.

$$\begin{aligned} E_N &= A_\Sigma q(F + \dot{q}^2/N^2), \\ E_\eta &= A_\Sigma N q \partial_\eta F, \qquad E_V = A_\Sigma N q \partial_V F, \\ E_q &= A_\Sigma \left[N(F + q\partial_q F) + \frac{\dot{q}^2}{N} + \frac{2q\ddot{q}}{N} - \frac{2q\dot{q}\dot{N}}{N^2} \right]. \end{aligned} \tag{V-01}$$

Substituting the auxiliary solution into (V-01) yields (H-04). Under a time displacement ξ, the fields transform as $\delta q = \xi\dot{q}$, $\delta\eta = \xi\dot{\eta}$, $\delta V = \xi\dot{V}$ and $\delta N = (\xi N)^{\cdot}$. Integrating the $\dot{\xi}$ term gives the Noether identity (H-05). Fixing $N = 1$ before variation would lose E_N, so the gauge is selected afterwards.

The endpoint one-form of the GHY-completed action is the following. It directly gives both the fixed data in (B-06) and the momentum shifts in (B-07).

$$\Theta = [p_q \delta q + \delta(B_{\mathrm{EC}} + B_{\mathrm{NY}})]_i^f \,. \tag{V-02}$$

For the full connection equation, invert the epsilon contraction in (E-01) to obtain $e^a \wedge T^b - e^b \wedge T^a = 0$. Let ι_a denote contraction with the dual frame and set $t = \sum_a \iota_a T^a$. Contracting and summing over a gives $T^b = -e^b \wedge t$ in four dimensions. Contracting once more over b gives $t = -3t$, hence $t = 0$ and $T^b = 0$. No two-amplitude restriction was used, so omitted torsion components do not remain as unconstrained solutions.

Finally, setting $\eta = V = 0$ in the LC curvature blocks gives $\mathrm{Ric}_{00} = -3(DH + H^2)$ and $\mathrm{Ric}_{ii} = DH + 3H^2 + (k_j + k_k)/q^2$. Subtracting the scalar trace proves (E-02). A reduced trajectory has $D^2 q = 0$ and $(Dq)^2 = -\chi$, so the temporal component and spatial trace vanish, leaving exactly Ξ.

C.1 Irregular torus constraint and endpoints

After auxiliary elimination on the torus, $C = -p_q^2/(4A_\Sigma q)$ has $dC = 0$ and vanishing Hamiltonian vector field on its zero set. The equation $p_q = 0$ defines the same set but generates a nonzero q translation, so it is not the same constrained gauge system. The invertible coordinate $\mu = q^{3/2}$, $q > 0$, gives

$$L = -\frac{4A_\Sigma}{9N}\dot{\mu}^2, \qquad p_\mu = -\frac{8A_\Sigma}{9N}\dot{\mu}, \qquad C = -\frac{9p_\mu^2}{16A_\Sigma}. \tag{H-07}$$

The irregularity persists: a first-order affine $\mu = a_\mu + b_\mu \tau$ still requires $b_\mu = 0$ at second order. The same argument holds for any positive background lapse $N_0(t)$ with $\tau = \int N_0 dt$. An endpoint-fixing ξ gives $\delta N = d(\xi N_0)/dt$, hence zero integral. Conversely, any zero-integral lapse variation gives $\xi = N_0^{-1} \int_{t_i}^t \delta N ds$. These gauge directions exhaust lapse variations at fixed T, but do not remove the q_0 modulus. The two-form pulled back to a fixed-time slice of the static solution family is $dq_0 \wedge dp_q = 0$; this degeneracy does not identify the volume modulus as gauge.

C.2 Endpoint generator and constraints

$$
\begin{aligned}
B_{\rm end} &= -\frac{\mathcal{V}_\Sigma q^3 V}{\kappa^2} + 6\theta_{\rm NY}\mathcal{V}_\Sigma q^2\eta,\\
P_i &= p_i + \partial_i B_{\rm end}, \qquad \widetilde{p}_i = P_i - \partial_i B_{\rm end}, \quad Q^i = (q, N, \eta, V),\\
C' &= -\frac{\widetilde{p}_q^2}{4A_\Sigma q} - A_\Sigma q(\chi - \eta^2 + q^2V^2/9).
\end{aligned}
\tag{K-01}
$$

This coordinate-dependent generator is defined after GHY completion; it excludes the velocity-dependent EH endpoint term before completion. Since $P_i\,dQ^i = p_i\,dQ^i + dB_{\rm end}$, the two-forms agree. Symmetry of the Hessian gives $\{Q^i, \widetilde{p}_j\} = \delta^i_j$ and $\{\widetilde{p}_i, \widetilde{p}_j\} = 0$. The primary constraints are $\widetilde{p}_N, \widetilde{p}_\eta, \widetilde{p}_V$ and the secondary constraints are C', η, V, with nonzero factors removed from the latter two for $N, q > 0$.

$$
\begin{aligned}
\{\zeta_\alpha, \zeta_\beta\} &= \begin{pmatrix} 0 & 0 & -1 & 0\\ 0 & 0 & 0 & -1\\ 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \zeta = (\widetilde{p}_\eta, \widetilde{p}_V, \eta, V),\\
\det\{\zeta_\alpha, \zeta_\beta\} &= 1, \qquad P_\eta = 6\theta_{\rm NY}\mathcal{V}_\Sigma q^2, \quad P_V = -\mathcal{V}_\Sigma q^3/\kappa^2,\\
\Omega_{\rm red} &= dq \wedge dP_q + dN \wedge dP_N, \quad C_{\rm red} = -P_q^2/(4A_\Sigma q) - A_\Sigma q\chi.
\end{aligned}
\tag{K-02}
$$

On the second-class surface, $\eta = V = 0$ and $P_q = p_q$. Pulling back the two-form with $d\eta = dV = 0$ immediately gives (K-02). At $\chi = 0, P_q = 0$, $dC_{\rm red}$ remains zero. Fixed q, η, V suffice for $\delta B_{\rm end} = 0$ without fixing lapse. Momentum endpoint conditions must instead be mapped through the translation. Changing the constant-NY endpoint representative introduces neither new bulk degrees of freedom nor regularization of the torus constraint.

D Homogeneous shape variations and local vacuum evolution

D.1 General matrix action and five-direction variation

Eliminate torsion by the independent connection equation and use the GHY-completed LC action. In a fixed class-A coframe, $e = qe^{\mathsf{B}}\sigma$ has spatial curvature $f(n_{\mathsf{B}})/q^2$ and tracefree extrinsic curvature S, with $D = N^{-1}d/dt$.

$$
\begin{aligned}
n_{\mathsf{B}} &= e^{\mathsf{B}} n e^{\mathsf{B}}, \qquad \mathsf{S} = \tfrac{1}{2}e^{-\mathsf{B}}D(e^{2\mathsf{B}})e^{-\mathsf{B}},\\
L &= \frac{\mathcal{V}_\Sigma N q^3}{2\kappa^2}\left[-6H^2 + \mathrm{tr}(\mathsf{S}^2) + \frac{f(n_{\mathsf{B}})}{q^2}\right].
\end{aligned}
\tag{S-04}
$$

The formula for f is the spatial Ricci trace obtained from the Koszul connection in diagonal n, and extends by constant rotation to every real symmetric n. Counting both $\mathsf{B}, D\mathsf{B}$ as first order, expand the exponentials preserving noncommuting product order. Then $\mathsf{S} = D\mathsf{B} + O(3)$: quadratic commutator terms cancel between the two factors $e^{-\mathsf{B}}$. The potential is

$$
\begin{aligned}
f(n_{\mathsf{B}}) &= 6\chi - 2\mathrm{tr}(X\mathsf{B}) + U_2 + O(3),\\
U_2 &= 2[\mathrm{tr}(\mathsf{B}n)]^2 + 2(\mathrm{tr}n)\mathrm{tr}(\mathsf{B}^2 n)\\
&\quad - 4\mathrm{tr}(\mathsf{B}n\mathsf{B}n) - 4\mathrm{tr}(\mathsf{B}^2 n^2).
\end{aligned}
\tag{S-05}
$$

Indeed, insert $n_{\mathsf{B}} = n + \mathsf{B}n + n\mathsf{B} + \mathsf{B}^2 n/2 + n\mathsf{B}^2/2 + \mathsf{B}n\mathsf{B} + O(3)$ into f to obtain each term. Tracefreeness gives (E-04). The five directions are $\mathsf{B}_{11} = \mathsf{b}_1, \mathsf{B}_{22} = \mathsf{b}_2, \mathsf{B}_{33} = -\mathsf{b}_1 - \mathsf{b}_2$ and three

independent off-diagonal components. Their velocity Gram form is $\dot{\mathsf{b}}_1^2 + \dot{\mathsf{b}}_2^2 + (\dot{\mathsf{b}}_1 + \dot{\mathsf{b}}_2)^2 + 2(\dot{\mathsf{b}}_3^2 + \dot{\mathsf{b}}_4^2 + \dot{\mathsf{b}}_5^2)$, which is positive definite.

Diagonalize n to classify $X = 0$. The difference $X_i - X_j = (n_i - n_j)[2(n_i + n_j) - s_1]$ excludes three distinct eigenvalues. With two equal eigenvalues, write $n = (a, a, b)$; the difference is proportional to $b(b - a)$, so $b = a$ or $b = 0$. Substitution into (E-04) verifies both branches, proving necessity and sufficiency in (E-05).

Independently evaluating the momentum equation about the common-scale background gives $\delta G_{01} = (n_2 - n_3)D\mathsf{B}_{23}/q$ and cyclic permutations. General off-diagonal shape velocities are therefore not unrestricted. All momentum equations vanish for $n = 0$, and are satisfied in the diagonal class-A sector with zero off-diagonal velocities.

D.2 Torus lapse constraint and exact family

For $n = 0$ the potential vanishes identically. Substitute $q = q_0 e^{\epsilon\rho}$, $\mathsf{B} = \epsilon\mathsf{B}_1$, $N = N_0 + \epsilon\delta N$ into (S-04). The quadratic term is (S-01), while the cubic term and original lapse equation give

$$\begin{aligned} L^{[3]} &= \frac{A_\Sigma q_0^3}{6N_0}\left(3\rho - \frac{\delta N}{N_0}\right)[\mathrm{tr}(\dot{\mathsf{B}}_1^2) - 6\dot{\rho}^2], \\ E_N^{[2]} &= \frac{A_\Sigma q_0^3}{6N_0^2}[6\dot{\rho}^2 - \mathrm{tr}(\dot{\mathsf{B}}_1^2)] = \partial_{\delta N} L^{[3]}. \end{aligned} \tag{S-06}$$

Absence of the first-order lapse variation from the quadratic action does not remove the constraint. Vary the original lapse first, or vary the cubic action with respect to δN, to obtain the same second-order constraint. The quadratic Euler equations are $D^2\rho = D^2\mathsf{B}_1 = 0$, giving the affine solutions and cone in (S-01) as necessary conditions.

For sufficiency use (S-02). Before the constant congruence, mixed extrinsic curvature is $K = (3\epsilon h/\psi)\mathsf{P}$. Spatial Ricci vanishes and $DK + (\mathrm{tr}K)K = 0$. The temporal Ricci component is

$$\mathrm{Ric}_{00} = \frac{9\epsilon^2 h^2}{\psi^2}[1 - \mathrm{tr}(\mathsf{P}^2)] = -\frac{\epsilon^2}{\psi^2}[\mathrm{tr}(\Sigma^2) - 6h^2]. \tag{S-07}$$

A constant congruence preserves local Ricci flatness, and every constant positive definite spatial metric descends to the same lattice. This does not declare a lattice transformation to be gauge. First-order matrix expansion gives $g_\epsilon = q_0^2[I + 2\epsilon(\rho_0 I + \mathsf{B}_0 + h\tau I + \Sigma\tau)] + O(\epsilon^2)$, even if initial shape and Σ do not commute. The determinant gives q_ϵ in (S-02). For $h = 0$, positivity forces $\Sigma = 0$, realized by a static family.

Any first-order lapse variation is realized by choosing positive $N_\epsilon = N_0 + \epsilon\delta N + O(\epsilon^2)$ and composing the exact family with $\tau_\epsilon = \int N_\epsilon dt$. Since the background is static, this composition leaves its first-order spatial metric tangent unchanged. Local or initial-data comparisons differ from a two-endpoint fixed-metric variation problem; in the latter all affine scale and shape coefficients vanish.

D.3 Diagonal closure and short-time coefficients

Use diagonal n and $e^i = qe^{\beta_i}\sigma^i$, $\sum_i \beta_i = 0$. Put $m_i = n_i e^{2\beta_i}/q$ and evaluate f, X_i below on physical m. The Koszul connection gives the diagonal spatial Ricci eigenvalues $2m_i^2 - (\sum m)m_i + f$, whose tracefree parts are $X_i = 2m_i^2 - (\sum m)m_i + 2f/3$. Extrinsic curvature is diagonal, and the momentum and off-diagonal Einstein equations vanish, so this sector closes.

$$\begin{aligned} Dq &= qH, & DH &= -3H^2 - f/3, \\ D^2\beta_i &= -3HD\beta_i - X_i(m), & Dm_i &= m_i(2D\beta_i - H), \\ \mathcal{C} &= f + 6H^2 - \sum_i (D\beta_i)^2 = 0, & D\mathcal{C} &= -6H\mathcal{C}. \end{aligned} \tag{S-08}$$

Constraint propagation follows by differentiating (S-08) and using $Df = -2Hf - 2\sum_i X_i D\beta_i$. On the constraint, $D^2q = -q\sum_i (D\beta_i)^2/3$. At $q = q_0, \beta = D\beta = 0, H_0^2 = -\chi/q_0^2$, one has $D^2\beta_i = -X_i(n)/q_0^2$ and $DX_i(m) = -2H_0X_i(n)/q_0^2$, hence $D^3\beta_i = 5H_0X_i(n)/q_0^2$. Also $D^2q = D^3q = 0$ and

$$D^4q\big|_0 = -\frac{2q_0}{3}\sum_i (D^2\beta_i|_0)^2 = -\frac{2J_2(n)}{3q_0^3}. \tag{S-09}$$

The supplement compares 12 points at $q_0 = 1$, Nil and Sol, both signs of H_0, and $\hat{\tau} = 0.01, 0.02, 0.04$, using DOP853 with rtol 2×10^{-13}, atol 2×10^{-15} and maximum step 0.002. Constraint residuals and Taylor differences are saved. These point comparisons and local Taylor orders are distinct from a certified interval error bound.

E Supplementary material

The accompanying `script`, `data` and `LaTeX` folders contain curvature components, action, constraints and endpoints; shape variations and the exact torus family; quadratic and cubic invariants and the Ricci inverse problem; short-time coefficients and 12 numerical comparisons; global frames; canonical correspondence; and the English source. Bilingual instructions distinguish saved-data-only `render` from explicit `recompute` into a new output directory. The definitions and proofs are in the main text and Appendices A–D; running the code is not a prerequisite for the argument. The supplementary comparison guide records convention conversions.

F Revision note

This version uses geometric torsion, complete NY normalization, the physical spherical radius and fixed compact quotients. The GHY sign, auxiliary momentum equations, internal-dual metric factors and volume normalizations of the first version are corrected. The five-parameter Jacobi condition and the class-A classification of Sol are made explicit. Conversions from the earlier implementation are given in the supplementary convention guide; the changelog records the version changes.

The earlier NY-dependent bulk mixing and conditional MX classification, universal dynamic Pontryagin zero, Lorentz-scalar interpretation of the unweighted internal sum, and Higgsing/response correspondence are withdrawn. The auxiliary solution, reduced trajectory classification and static $C = -9\chi$ relation are retained in the domains specified in Sections 4–6.

This version also develops time-dependent class-A curvature formulas, relative Pontryagin endpoint quantities, the full connection equation and the tracefree coframe residual on reduced trajectories. It adds homogeneous shape variation, exact realization of mixed torus tangents, short-time Nil and Sol coefficients, selected quadratic and cubic contractions with the Ricci inverse problem, global frames of the static torus and canonical endpoint correspondence. The main text and appendices specify the variation domains and boundary conditions of each conclusion.

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